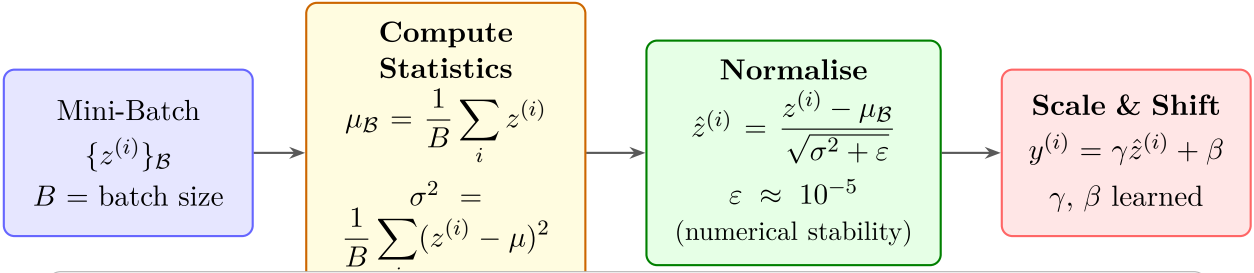

Why Batch Normalisation Helps

1. Reduces *internal covariate shift* — keeps layer inputs in a stable distribution.
2. Allows **higher learning rates** without divergence.
3. Acts as a **regulariser**, reducing reliance on dropout.
4. At **inference**: use running mean $\bar{\mu}$ and running variance $\bar{\sigma}^2$ (accumulated during training).

$\gamma = 1,\ \beta = 0$ *initially — model learns to undo normalisation if beneficial.*